



2807 West County Road 75
Monticello, MN 55362

November 13, 2023

L-MT-23-048
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant Licensee Event Report 2023-002-00

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-263/2023-002-00 per 10 CFR 50.73(a)(2)(iv)(A).

If you have any questions about this submittal, please contact Carrie Seipp, Senior Regulatory Engineer, at 612-330-5576.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Greg D Brown', written over a light blue horizontal line.

Greg D Brown
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-263/2023-002-00**

3 pages follow



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Monticello Nuclear Generating Plant	☒ 050 ☐ 052	2. Docket Number 263	3. Page 1 OF 3
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4. Title
Reactor Scram and Containment Isolation due to Valve Responses during Main Turbine Control Valve Testing

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
09	27	23	2023	- 002 -	00	11	13	23	Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 075
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Carrie Seipp, Senior Nuclear Regulatory Engineer	Phone Number (Include area code) 612-330-5576
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	TA	FCV	G080	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year
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16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On September 27, 2023 at 1041 CDT, with Monticello Nuclear Generating Plant in Mode 1 at approximately 75 percent power for the semiannual Turbine Control Valve (CV) test, an unplanned Reactor Scram, Group I Primary Containment Isolation, and Group II Primary Containment Isolation occurred.

The cause of the Reactor Scram was exceeding the Reactor Pressure Vessel (RPV) high pressure scram setpoint during CV testing. CV-1, CV-2 and CV-3 percentage open did not increase enough when CV-4 was closed for testing. The reduction in steam flow caused the RPV pressure to exceed the scram setpoint. Subsequently, an expected low RPV level caused the Group II Primary Containment Isolation. The cause of the Group I Primary Containment Isolation was a transient in the steam line high flow instrumentation sensing lines for the C Main Steam Line.

Corrective actions included inspection and adjustment of the four CVs to within manufacturer's specifications. Channel calibration and functional tests were performed to verify that all C Main Steam Line flow instruments met requirements.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Monticello Nuclear Generating Plant	☒ 050	2. DOCKET NUMBER 263	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER 002	REV NO. 00

NARRATIVE**EVENT DESCRIPTION**

On September 27, 2023 Monticello Nuclear Generating Plant (MNGP) was operating in Mode 1 and approximately 75 percent power for the semiannual Turbine-Generator Operational Tests. At 1041 Central Daylight Time (CDT), while cycling closed the Turbine Control Valve 4 (CV-4) per the Turbine CV Test, an unplanned Reactor Scram, Group I Primary Containment Isolation, and Group II Primary Containment Isolation occurred.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) for an event or condition that resulted in the automatic actuation of the Reactor Protection System (RPS) [EIS CODE: JC] including reactor scram and containment isolation signals affecting containment isolation valves in more than one system and multiple Main Steam Isolation Valves (MSIVs) [EIS CODE: ISV].

EVENT ANALYSIS

The Turbine Control Valves (CVs) [EIS CODE: FCV] are a part of the MNGP Main Turbine System [EIS CODE: TA]. The CVs control reactor pressure, in conjunction with the turbine bypass valves, by adjusting position to control steam flow.

During the semiannual Turbine-Generator Operational Tests, reactor power is reduced to approximately 75 percent to allow for testing of each of the four CVs. While one CV is stroked closed, the other three CVs are expected to increase percent open to compensate for the increased reactor pressure. While stroking CV-4 closed per the procedure, the percent open of CV-1 and CV-2 initially increased then stalled and CV-3 percent open did not change. In turn, total steam flow through the CVs lowered and the reactor pressure exceeded the (RPV) high pressure scram setpoint, resulting in an automatic Reactor Scram. The RPV water level lowered to below the Group II isolation setpoint as expected, resulting in a Group II Primary Containment Isolation. Additionally, a Main Steam Line high flow condition was sensed in C Main Steam Line, resulting in a Group I Primary Containment Isolation and closure of the Main Steam Isolation Valves. There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

ASSESSMENT OF SAFETY CONSEQUENCES

All safety systems functioned properly during the transient. All control rods fully inserted during the scram. The High Pressure Coolant Injection (HPCI) system was placed in service to control RPV pressure, but HPCI did not inject into the RPV and was not needed to control RPV water level.

There were no radiological, environmental, or industrial impacts associated with this event. The health and safety of the public and site personnel were not impacted during this event.

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NARRATIVE**CAUSE OF THE EVENT**

The cause of the RPV high pressure scram setpoint being exceeded was attributed to performance of Control Valve testing with the operating CVs just below the transition point between the large outer piston and the smaller inner piston of the actuator. The transition requires an increased hydraulic pressure to continue moving the CV in the open direction. In the test configuration, this can result in the CV stalling at the transition point. The transition point can occur at a slightly different percentage open for each valve based on control valve actuator settings, linkage tolerances, and valve wear. During troubleshooting of the event, it was identified that maintenance performed on the CVs in April 2023 changed the transition points for the valves which affected how the valves respond to system demand at test conditions. Troubleshooting identified that only CV-4 was above the transition point prior to the close stroke test of the other CVs. That is, CV-4 did not move through the transition point and performed the largest percentage open adjustments to control reactor pressure during close stroke testing of CV-1, CV-2, and CV-3. However, CV-1, CV-2, and CV-3 were operating just below their transition points during testing. When CV-4 was stroked closed, CV-1, CV-2, and CV-3 were not able to overcome the transition point to increase the percent open enough, and without the increased steam flow, the RPV pressure exceeded the Reactor Scram setpoint. As expected, the RPV water level lowered to below the Group II isolation setpoint, resulting in a Group II Primary Containment Isolation.

The cause of the Group I Primary Containment Isolation was a sensed, rather than actual, high steam line flow on the C Main Steam Line. Plant process computer data showed a transient on the C Main Steam Line flow transmitter. This transient was sensed as high flow by the Group I isolation high steam flow logic. No actual steam leak existed as determined via redundant indications and system walkdown.

CORRECTIVE ACTIONS

CV-1, CV-2, CV-3, and CV-4 were inspected and adjusted to the manufacturer's recommended control valve actuator settings, including adjustment of the operating cylinder closed end overtravel and control valve cracking points. Hysteresis testing confirmed that actuator transition points were within manufacturer's specifications. The valves were retested satisfactorily.

Channel calibration and functional tests were performed to verify that all C Main Steam Line flow instruments met requirements. No adjustments were needed.

PREVIOUS SIMILAR EVENTS

No previous similar events have occurred at MNGP in the prior 3 years.